SCIENCE

Subjective Test-02 [2026-2027]

Class-IX

CBSE

Max. Marks: 40 **Time: 90 Minutes**

DATE: 05 / 07 / 2026

Topics Covered

Biology: Cell: The building block of life :
Introduction : Introduction, How to study cells? Discovery Of Cell, Microscopy, magnification, Diversity Of Cell
Plasma Membrane And Osmosis, Types Of Solution And Cell Wall, Cell Organelles : Nucleus, Difference Between Eukaryotic And Prokaryotic Cell, Cell Organelles : Endoplasmic Reticulum, Golgi Apparatus, Lysosomes, Mitochondria, Plastids And Its Types, Vacuoles, Cellular Division And Cancer : Cell Division And Its Types, Cancer, Cell theory

Chemistry: Exploring Mixtures and their Separation: Chemical Classification Of Matter, Introduction to Matter ; Understanding Classification of Matter based on Physical and Chemical properties- Pure substances (Elements and Compounds) and Mixtures, Mixtures And Their Classification ; True Solutions And Their Properties, Definition of Mixtures ; Classification of Mixtures - Homogeneous and Hetereogeneous Mixtures.

1. Binary Solutions- Solute and Solvent
2. Properties of Solution
3. Classification of Solution - Alloys ; Solid Solution ; Liquid Solution ; Gaseous Solution, Suspensions And Colloids, Properties of Colloidal Solution - Tyndall Effect, Types of Colloids- (Based on Dispersion Medium), Other Properties of Colloidal Solution, Properties of Suspensions Difference between True Solutions , Suspensions and Colloids

Understanding Concentration Of Solution And Its Ways Of Expression :

1. Classification of Solution
2. Concentration Terms, W/W %- (mass by mass percentage of a solution), W/V % -(mass by volume percentage of solution), V/V %- (volume by volume percentage of solution)

Physics: Basic physics : Quantities, Fundamental Quantities, Units, Scalars And Vectors And Decimal Notation Describing Motion Aorund Us : Motion — distance & displacement, Problems on distance and displacement, velocity, acceleration, Acceleration, Uniform & Non-uniform Acceleration, Numerical Problems

Name of the Candidate (in Capitals): _____

Roll Number: _____

Candidate's Signature: _____ Invigilator's Signature: _____

General Instructions:

Read the instructions carefully and follow them:

1. ***This question paper comprises 19 questions. All questions are compulsory.***
2. ***The question paper contains THREE sections – Section – A, Section – B & Section C***
3. ***Attempt questions based on specific instructions for each Part. Write the correct question number and part thereof in your answer sheet.***
4. ***Separate instructions are given with each question/part, wherever necessary.***
5. ***Adhere to the prescribed word limit while answering the questions.***

SECTION-A [BIOLOGY]

Multiple Choice Questions

1.	Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cell are: (A) Centrosomes (B) Ribosomes (C) Lysosomes (D) Mitochondria	**1**
2.	A cell organelle containing hydrolytic enzymes is (A) Lysosome (B) Microsome (C) Ribosome (D) Mesosome	**1**
3.	Which plastids are colourless? (A) Chromoplasts (B) Chloroplast (C) Leucoplasts (D) None of the above	**1**
4.	**Assertion (A):** Cell wall is a non-living part of the cell. **Reason (R):** It offers protection, definite shape and support. Choose the correct option: (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is not the correct explanation of A. (C) A is true, but R is false. (D) A is false, but R is true.	**1**
5.	Give reasons – (i) Why does the skin of your finger shrink when you wash clothes for a long time? [1 M] (ii) Where do the lipids and proteins constituting the cell membrane get synthesized? [2 M]	**3**
6.	Based on the number of cells, living organisms are classified as unicellular and multicellular. (i) Name two unicellular organisms. [1 M] (ii) What is meant by the division of labour in multicellular organisms? [1 M] (iii) Name one prokaryotic and one eukaryotic unicellular organism. [1 M] (iv) "Every multicellular organism has come from a single cell." Justify this statement. [1 M] (v) Write one common feature between an *Amoeba* and a white blood cell of humans. [1 M]	**5**
7.	Sahil was reading about cell organelles and imagined a cell as a tiny living factory. His teacher explained that just like a factory has different departments - a power unit, a packaging department, a cleaning unit, and a control room - a cell also has different organelles each performing a specific job. The teacher described one organelle as the 'powerhouse of the cell' because it supplies energy for most cellular activities. Another organelle was described as the 'packaging and shipping centre' that modifies, sorts, and packages proteins into vesicles for transport or secretion. A third organelle was called the 'clean-up system' of the cell because it breaks down unwanted proteins, carbohydrates, fats, and damaged parts of the cell. (i) Name the 'packaging and shipping centre' of the cell. Describe its structure and explain how it is functionally linked to other cell organelles.	**2**

SECTION-B

CHEMISTRY

Multiple Choice Questions

8.	Why does a suspension appear opaque or turbid, whereas a colloid appears translucent? (A) Suspension particles completely absorb all light, while colloids reflect it. (B) Suspension particles are large enough they block and reflect almost all incident light, whereas smaller colloidal particles scatter light in patterns. (C) Colloidal particles are completely dissolved, whereas suspensions are not. (D) Suspensions are homogeneous mixtures.	**1**
9.	Which of the following concentration terms does not change with a change in temperature of the solution? (A) Mass by mass percentage (B) Volume by volume percentage (C) Mass by volume percentage (D) Both (B) and (C)	**1**
10.	**Assertion:** An alloy like duralumin is chemically classified as a mixture. **Reason:** Alloys show the properties of their constituent elements and can vary in composition without a fixed chemical formula. (A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion. (B) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion. (C) Assertion is true but Reason is false. (D) Assertion is false but Reason is true.	**1**

Short Answer Type Questions

11.	A teacher gives three unmarked beakers A, B, and C to a group of students. Beaker A contains a copper sulphate solution, Beaker B contains a mixture of chalk powder and water, and Beaker C contains milk. Explain how the students can use the following criteria to identify each mixture. (i) Path of a beam of light [1 M] (ii) Stability upon standing [1 M] (iii) Filtration using ordinary filter paper [1 M]	**3**

Long Answer Type Questions

12.	(i) Classify the following as elements, compounds, or mixtures. Provide one characteristic reason for your classification in each case: [3 M] (a) Carbon dioxide gas (b) 22-carat gold (c) Diamond (ii) A pharmacist prepares a cough syrup by dissolving 12 grams of active medicinal ingredient in enough water to make total final volume of 240 ml of syrup. Calculate the % w/v concentration of medicine. [2 M]	**5**

Case Based Questions

13.	Anant wants to make Gulab Jamun syrup at home. He adds 220 g of sugar to 100 mL of water at room temperature (20°C-25°C) and stirs thoroughly. After some time, he notices that a few sugar crystals stop dissolving and settle at the bottom of the vessel, no matter how much he stirs. His elder brother advises him to place the vessel on a gas stove and heat it up. (i) What type of mixture is formed when no more sugar can be dissolved in water at a given, fixed temperature? [1 M] (A) Unsaturated solution (B) Saturated solution (C) Heterogeneous mixture (D) Supersaturated solution (ii) How does temperature dependence of sugar solubility be compared to the solubility of a solute like carbon dioxide in water? [1 M] (A) The solubility of solid sugar increases with temperature, whereas the solubility of Carbon dioxide decreases with temperature. (B) Temperature changes have absolutely no effect on either solid or gas solubility. (C) Sugar solubility decreases with heat, while solubility of Carbon dioxide increases with heat. (D) Both sugar and carbon dioxide become more soluble as temperature increases.	**2**

SECTION-C

PHYSICS

Multiple Choice Questions

14.	Which of the following physical quantity is vector? (A) Speed (B) Distance (C) Energy (D) Average velocity	**1**
15.	The numerical ratio of displacement to distance for a moving object is always: (A) Always less than 1 (B) Always equal to 1 (C) Always more than 1 (D) Equal to or less than 1	**1**
16.	**Assertion (A):** An object can have zero displacement but non-zero distance. **Reason (R):** Distance depends on the actual path travelled whereas displacement depends only on initial and final positions. (A) Both A and R are true and R is the correct explanation of A. (B) Both A and R are true but R is not the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true.	**1**

Short Answer Type Questions		
17.	A person walks from A to B at 5 km/h. While returning from B to A along the same path, it starts raining, and the person walks at 3 km/h. Find his average speed and average velocity.	**3**
Long Answer Type Questions		
18.	(i) Define Retardation. A car accelerates uniformly from rest to a speed of 30 m/s in 10 seconds. Calculate the acceleration of the car. [3 M] (ii) Differentiate between scalar and vector quantities with examples. [2 M]	**5**
Case Based Questions		
19.	Rohan participates in a school sports event. During practice, he starts from the school gate and walks 50 m towards East. He then turns and walks 30 m towards North. After taking a short break, he walks 50 m towards West and reaches a point directly north of the school gate. While discussing his movement with his friends, he says that he has travelled a long distance but has not moved very far from his starting point. His teacher asks the class to calculate both the distance travelled and the displacement of Rohan. Based on the above information, answer the following questions: (i) What is the total distance travelled by Rohan? [1 M] (ii) What is the magnitude and direction of his displacement from the starting point? [1 M]	**2**